# FLAIR

**F**ederated **L**earning **A**ssessment for **I**CU **R**esearch

## The Federated External-Validation Network for ICU AI on the Common Longitudinal ICU Format (CLIF)

CLIF Consortium · FLAIR Working Group

July 6, 2026

MIMIC-CLIF · Rush · Northwestern Medicine · UChicago Medicine ▪ 278,065 ICU stays

# ICU AI has a trust bottleneck

- Models for sepsis, extubation, mortality, and operations now appear **faster than evidence can validate them**
- A strong internal AUC does not answer the question a hospital actually has
- The hard part is not training another model; it is proving whether a model works **somewhere it was not built**

**The clinical-scientific question**

Before we trust or deploy this model, can it survive contact with our ICU population, our workflows, and our data?

**FLAIR turns that question into a repeatable network test.**

# The one-line value proposition

Develop models on public MIMIC-CLIF.

Validate them across independent hospitals.

Keep patient-level data inside each institution.

**The model goes to the data. Only aggregate, cell-suppressed reports come back.**

# The result that makes this urgent

| Site | Sepsis AUROC |
|---|---|
| MIMIC-CLIF internal | **0.697** |
| Rush | **0.430** |
| Northwestern Medicine | **0.449** |
| UChicago Medicine | **0.440** |

**What FLAIR surfaced**

A model that looked useful on public ICU data became **worse than chance**.

The highest-stakes task failed at every external health system.

This is not a one-hospital anomaly. It is the exact failure mode external validation must reveal before clinical claims are made.

# Why FLAIR

# Public benchmarks are necessary, but not sufficient

| Dimension | Public ICU benchmarks | FLAIR |
|---|---|---|
| Test data | centralized / single-site | **federated**; stays at each hospital |
| Main value | reproducible development | **external validation** under real institutional variation |
| Tasks | mortality, LOS, AKI, generic sepsis | deployment-facing mortality, LTACH, extubation, CDC-ASE sepsis, readmission |
| Split | often global or random | **temporal** at external sites; pseudo-prospective |
| Scoring | frequently row-level | episodic / landmark / peak; closer to operational use |
| Saturation risk | public test data can be tuned against | private site data never enter public hands |

FLAIR complements MIMIC, eICU, HiRID, YAIB, EHRSHOT, and MEDS. It adds the missing federated validation layer.

van de Water et al. 2023; Wornow et al. 2023; Arnrich et al. 2025

# External validation is too slow today

- Independent EHR model validation usually requires one-off data-use agreements, local analyst time, and repeated reimplementation
- Flagship systems have taken **years** to reach only a handful of sites
- The Epic Sepsis Model fell to **AUC 0.63** in independent evaluation, against vendor-claimed 0.76–0.83

**FLAIR changes the unit of work**

From bespoke bilateral studies to a standing, governed model-to-data protocol over a shared ICU standard.

Wong et al., JAMA Intern Med 2021; Habib et al. 2021; Sendak et al. 2025; Adams et al. 2022

# The infrastructure gap: no standing ICU validation network

- **OHDSI / LEGEND** federates studies over OMOP, but not an ICU benchmark catalog
- **OpenSAFELY** moves analytic code to secure data, but not critical-care model benchmarking
- **MedPerf** and **FeTS** prove model-to-data and federated training in imaging
- **DREAM / EHR DREAM** show challenge-style model-to-data evaluation

**Missing for ICU EHR: CLIF standardization + reusable, extensible tasks + governed federated evaluation.**

Reps et al. 2020; OpenSAFELY 2021; Karargyris et al. 2023; Pati et al. 2022; Guinney et al. 2018; Bergquist et al. 2024

# What FLAIR Is

# FLAIR is three coupled products

**1. Data layer**

CLIF critical-care standard plus public MIMIC-CLIF entry point for development.

**2. Task layer**

Five inspectable Python tasks today; open, governed, PR-extensible catalog.

**3. Protocol layer**

Committee-reviewed model-to-data execution with aggregate-only reports.

**This is not another ICU model. It is the evidence layer models must pass through.**

# FLAIR is open infrastructure for ICU evaluation standards

- Open-source task definitions, feature contracts, report generation, and evaluation protocol
- A common way to define cohorts, prediction times, labels, splits, metrics, report bundles, and audit metadata
- New tasks are added as inspectable Python modules, not one-off private benchmark forks

**Task catalog standard**

Contribution requires MIMIC-CLIF reproducibility, clinician face-validity review, and multi-site governance sign-off.

**The goal is reusable, auditable, extensible ICU AI evaluation across the CLIF community.**

# What each stakeholder gets

| Stakeholder | FLAIR value |
|---|---|
| Model developers | A public development path plus credible external evidence without acquiring private ICU data |
| Hospitals | Local control: approve what runs, keep patient rows inside the institution, receive comparable performance reports |
| Journals / reviewers | Audit hashes, reproducible task code, cohort summaries, confidence intervals, calibration where meaningful |
| CLIF consortium | A high-value use case that makes CLIF adoption immediately useful for benchmarking and collaboration |
| Patients / public | Less reliance on claims from one development hospital before real-world evaluation |

# Why CLIF makes the network possible

- CLIF maps ICU EHR content into shared relational tables and mCIDE categories: vitals, labs, medications, respiratory support, microbiology, ADT, diagnoses, and more
- Task code refers to **shared table names and category values**, not local idiosyncratic EHR extracts
- MIMIC-CLIF expresses MIMIC-IV in CLIF v2.1 with approximately **72,800 ICU stays**
- Result: developers debug publicly, then run the same task/model contract across private CLIF sites

**CLIF is the portability layer. FLAIR is the validation layer.**

Rojas et al. 2025; Johnson et al. 2023

# Developers can plug in at the representation they need

| CLIF-native features | MEDS event stream | Wide table |
| --- | --- | --- |
| Direct table-based extraction for models that operate on CLIF relations. | Optional event-sequence export for MEDS-native pipelines. | YAIB-style matrix for classical ML and one-row-per-prediction workflows. |

**FLAIR standardizes the task and evaluation contract, not the model architecture.**

Model teams can use their preferred representation as long as outputs meet the prediction-file contract. FLAIR can assist with supported formats; the model owner controls feature engineering and modeling choices.

Arnrich et al. 2025; van de Water et al. 2023; Wornow et al. 2023

# The initial catalog covers decisions ICUs actually face

| Task | Outcome | Grid | Mode | Prev. |
|---|---|---|---|---|
| T1 | ICU daily mortality | daily 07:00 | landmark | 0.125 |
| T2 | LTACH discharge | daily 07:00 | landmark | 0.062 |
| T3 | Extubation failure 24 h | at extubation | episodic | 0.054 |
| T4 | Sepsis (CDC ASE) 6 h | hourly | peak | 0.061 |
| T5 | Unplanned ICU readmission | at ICU discharge | episodic | 0.093 |

- One Python module per task: `META` + `build()` returns one row per prediction opportunity
- Catalog acceptance requires MIMIC-CLIF reproducibility, clinician face validity, and at least two-site sign-off

# Sepsis is the stress test

- Task 4 uses CDC Adult Sepsis Event within 6 hours on an hourly, hospital-wide grid
- The prediction window ends strictly before onset: no future treatment signal
- Peak scoring collapses each stay to the maximum risk, matching a one-alarm operational interpretation
- It is rare, high-stakes, and sensitive to practice-pattern shift

Kamran et al., NEJM AI 2024; Rhee et al. 2019

**Why this matters**

If a sepsis model fails across hospitals, the field needs a safe way to find out before deployment, not after.

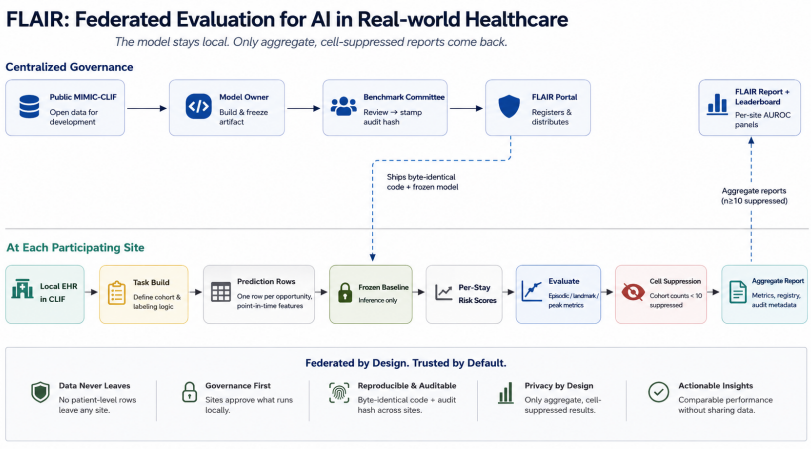
# How It Works

# Where FLAIR fits in the AI/CDSS lifecycle

| Step | In plain terms | Owner |
|---|---|---|
| Cohort + task | Decide who is included, when prediction happens, and what counts as the outcome | **FLAIR standard**; community-extensible |
| Feature engineering | Turn raw clinical history into model inputs using CLIF-native, MEDS, or wide data | Model owner; FLAIR assists with formats |
| Model development | Train, tune, document, and freeze the algorithm | Model owner |
| Federated evaluation | Run the frozen artifact locally at each site using the same task and metrics | **FLAIR protocol**; site-approved |
| Evidence report | Return comparable, aggregate, cell-suppressed performance summaries | **FLAIR report standard** |

**Encapsulation makes evaluation faster and cleaner: shared task logic, private models, repeatable reports.**

# The model-to-data loop




Central governance approves and stamps the artifact; each site runs it locally on CLIF; only aggregate, cell-suppressed reports return.

# Governance is part of the product

- Benchmark Committee reviews reproducibility, clinical face validity, task fit, and security
- Sites retain final approval over what runs locally
- Every report includes task hash, package version, split, report mode, n, prevalence, and threshold

**Privacy posture**

No patient-level data leave any site. Counts below 10 are suppressed; sparse subgroups and calibration bins are withheld.

**The network is usable because trust, review, and suppression are built in.**

# Comparability safeguards make results credible

1. **Stitched hospitalizations:** fragmented clinical episodes are joined into one `hospitalization_join_id`; one episode never straddles train/test
2. **Strict point-in-time features:** only data at or before `prediction_dttm` can be used
3. **Pseudo-prospective external splits:** default is oldest 75% train, newest 25% test at each site
4. **Audit hashes:** sites can prove they ran byte-identical task code and policy metadata

MIMIC-CLIF uses a deterministic hash split because de-identified dates do not preserve usable temporal order.

# Evaluation matches the way a task is used

| Mode | Prediction unit | Report bundle |
|---|---|---|
| Episodic (T3, T5) | one prediction per stay | AUROC, AUPRC, ROC/PR curves, calibration, decision curves, subgroup AUROC |
| Landmark (T1, T2) | one daily prediction while at risk | per-landmark AUC curve plus pooled dynamic AUC, lead-time and skill plots |
| Peak (T4) | stay collapsed to maximum risk | discrimination and subgroup metrics; calibration/DCA omitted to avoid peak-calibration distortion |

**The benchmark measures operational behavior, not just row-level discrimination.**

# The reference baseline is deliberately shippable

- One frozen XGBoost model over point-in-time count features, following EHRSHOT/FEMR tradition
- CLIF events map to a fixed 211-code Event List Format vocabulary
- Counts include only codes strictly before prediction time; units and discharge diagnoses are excluded
- Hyperparameters tuned once on MIMIC-CLIF; model and dictionary ship unchanged to sites for inference only

The baseline is not the product. It is the benchmark check that exposes transport behavior.

Wornow et al. 2023; Steinberg et al. 2021

# What We Learned

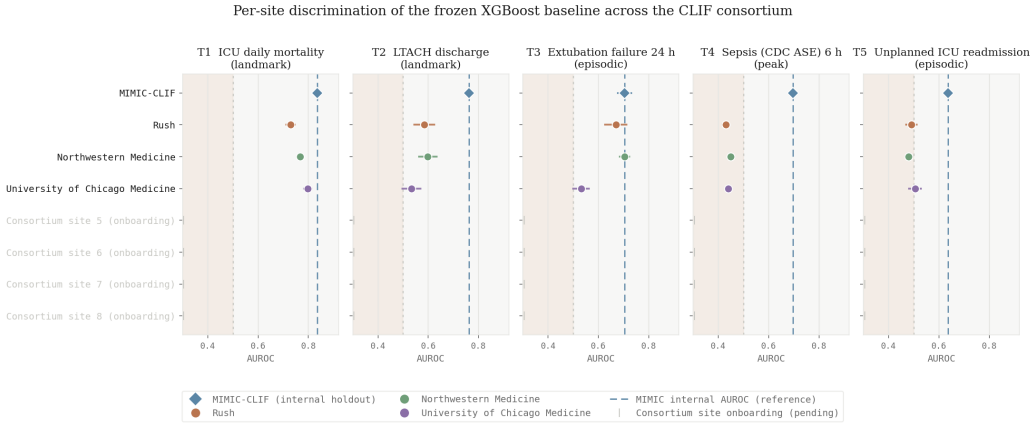
# Four reporting sites; four onboarding

| Site | ICU stays | F% | Age | SOFA |
|---|---|---|---|---|
| MIMIC-CLIF | 72,778 | 43.2 | 67 | 5 |
| Rush | 53,356 | 47.8 | 63 | 4 |
| Northwestern Medicine | 114,335 | 44.0 | 66 | 5 |
| UChicago Medicine | 37,596 | 43.5 | 61 | 6 |
| Consortium sites 5–8 | — | — | — | — |

**278,065 ICU stays across four independent reporting health systems.**

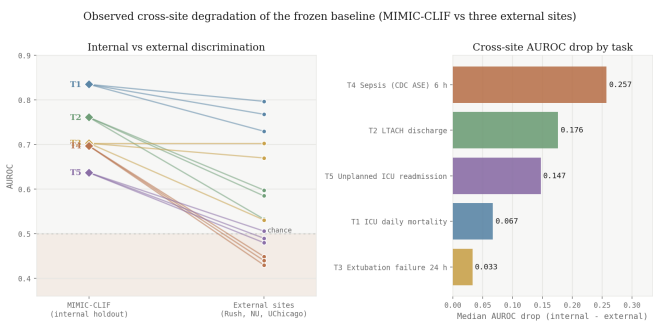
Placeholder rows leave space for the eight-site consortium without fabricating results.

# Per-site performance is the headline product




A single pooled AUROC would hide the story. FLAIR makes the per-site spread visible.

# Strong internal discrimination does not survive transport




Median internal → external AUROC drop:

- Sepsis **0.257**
- LTACH 0.176
- Readmission 0.147
- Mortality 0.067
- Extubation 0.033

**Task-dependent**, not uniform.

# Transportability is heterogeneous, not binary

**Transports**

**Mortality (T1)**
external 0.73–0.80
vs 0.835 internal

**Collapses**

**Sepsis (T4)**
below chance everywhere

**Readmission (T5)**
near chance

**Splits**

**Extubation (T3)**
Northwestern **0.703**
UChicago **0.531**

**This is why federated external validation matters: site effects are the signal.**

# Why It Wins

# FLAIR's defensible advantages

- **Saturation resistance:** evaluation data stay siloed and private, limiting tune-to-test and pretraining-on-test-set risks
- **Evaluation, not training:** aggregate metrics leak less than federated-learning updates; no per-site model object is shared
- **Auditability:** byte-identical code and hashes make labels, exclusions, splits, and cohorts re-inspectable
- **Extensibility:** new tasks can enter through PRs and multi-site governance instead of one-off benchmark forks
- **Adoption flywheel:** every CLIF site increases the value of every future model evaluation

# What FLAIR is not claiming

- It does not certify any model for bedside use
- It does not replace prospective silent trials or clinical-impact studies
- It does not move patient-level data or expose row-level errors centrally
- It currently covers structured EHR binary prediction tasks, not notes, waveforms, imaging, treatment effects, or generative outputs

**FLAIR is the pre-deployment evidence layer between local development and clinician-facing claims.**

# The ask

Join the federated ICU validation network.

Map to CLIF, run approved artifacts locally, and help define the next benchmark tasks.

| For sites | For developers | For funders |
| --- | --- | --- |
| Contribute private external evidence without sharing private rows. | Bring frozen models for auditable, multi-site validation. | Invest in infrastructure that scales evidence, not just algorithms. |

**Develop on public MIMIC-CLIF. Validate federated.**

# Takeaways

1. FLAIR fills the missing layer in ICU AI: external validation across real hospitals without patient-data movement
2. CLIF makes the code portable; FLAIR makes the evaluation governed, auditable, and reusable
3. The first observed results already justify the network: sepsis falls below chance at every external site, while mortality transports and extubation splits by hospital
4. The contribution is not a model; it is a reproducible way to ask whether ICU models behave alike across institutions

# Thank you

FLAIR — Federated Learning Assessment for ICU Research

github.com/clif-consortium/FLAIR

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# Backup

# Backup: pooled dynamic AUC for landmark tasks

For a stitched hospitalization $h$ with event onset $T_h$, discharge $D_h$, horizon $\tau$, at landmark time $t$:

Risk set:

$$R(t) = \{\, h : T_h > t,\ D_h > t \,\}$$

Dynamic label:

$$Y_h(t) = \mathbf{1}\{\, t < T_h \le t + \tau \,\}$$

Predicted risk:

$$p_h(t) = \hat{P}(t < T_h \le t + \tau \mid \mathcal{H}_h(t))$$

Heagerty & Zheng 2005; van Houwelingen 2007; Blanche et al. 2013

Tie kernel: $\psi(a, b) = \mathbf{1}\{a > b\} + \frac{1}{2}\mathbf{1}\{a = b\}$

$$\widehat{\mathrm{AUC}}(t) = \frac{\sum_{i,j \in R(t)} Y_i(t)\,[1 - Y_j(t)]\,\psi(p_i(t), p_j(t))}{\left(\sum_i Y_i(t)\right)\left(\sum_j [1 - Y_j(t)]\right)}$$

$$\mathrm{AUC}_{\text{pooled}} = \frac{\sum_t \widehat{\mathrm{AUC}}(t)\, w_t}{\sum_t w_t}, \quad w_t = \text{cases}_t \cdot \text{controls}_t$$